Deepfake Image Detection: Comparative Analysis of State-of-the-Art Methods

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Abstract

The rapid growth of generative artificial intelligence has made the creation of highly realistic synthetic images, or deepfakes, increasingly accessible. These developments pose significant challenges to digital media authenticity, security, and privacy. In this study, we conduct a comprehensive comparative analysis of eight state-of-the-art deepfake detection models, encompassing diverse architectural designs and methodological strategies, including Dense CNN, CSTAN, DDEM, ensemble frameworks, LEDNet, RAW-domain detection pipelines, and stochastic degradation-based approaches. All models are evaluated under a unified experimental setting using a recent and diverse dataset. Based on the comparative analysis, the most suitable model for deployment is selected and integrated into a practical software system for real-world validation. Experimental results highlight the trade-offs between detection accuracy, robustness to distortions, computational efficiency, and deployment performance, contributing to a more practical understanding of deepfake image detection in real-world scenarios.

Keywords: Deepfake detection, Image forensics, Convolutional neural networks, Generative adversarial networks, Multimedia security, Robust detection

1 Introduction

Advances in artificial intelligence and deep learning have significantly improved the ability to generate highly realistic synthetic images. Among these developments, deepfake technology has emerged as one of the most concerning because it can create manipulated images that are visually convincing and difficult for humans to identify. Deepfake images are commonly generated using powerful models such as Generative Adversarial Networks (GANs) and diffusion-based architectures, which are capable of learning complex visual patterns and producing highly realistic fake content. Although these technologies have useful applications in entertainment, media production, and digital creativity, their misuse has created serious risks for digital security, privacy, and public trust [1, 2].

The rapid spread of deepfake content has increased the need for reliable detection systems. Deepfake images can be used for misinformation, impersonation, identity theft, defamation, fraud, and other malicious activities. The growing realism of fake content makes manual detection increasingly difficult, especially when manipulated images are shared through social media and online platforms. Recent reports and case examples have shown that deepfake misuse is no longer a theoretical issue but a practical cybersecurity threat with social, financial, and ethical consequences. As deepfake generation methods continue to improve, detection approaches must also evolve to remain effective against modern manipulation techniques [2].

Many existing deepfake detection methods rely on deep learning models that analyze visual inconsistencies, texture artifacts, frequency patterns, and reconstruction errors. Convolutional Neural Networks (CNNs) have been widely used because of their strong ability to extract spatial features from images. More recent approaches have introduced attention mechanisms, diffusion-based reconstruction models, vision–language learning, ensemble strategies, and RAW-domain analysis to improve detection robustness and generalization. However, despite strong performance on benchmark datasets, many existing methods still face important limitations such as poor cross-dataset generalization, high computational cost, dependence on large pretrained models, and reduced effectiveness against newly emerging generative techniques. These limitations indicate that there is still a need for a more comprehensive comparative evaluation of different deepfake detection paradigms under consistent experimental conditions [3–6].

This study focuses on deepfake image detection using a comparative deep learning framework. A recent and diverse dataset of real and fake images is used to train and evaluate multiple deep learning models representing different detection strategies. In this research, eight models are implemented and compared, including D-CNN, CSTAN, OD-ResNet-34, DDEM, Ensemble Framework, LEDNet, RAW-SBI Pipeline, and SDAug. These models were selected because they represent a broad range of current approaches, including dense convolutional learning, attention-based methods, diffusion-based detection, multimodal learning, ensemble classification, sensor-level analysis, and robustness-oriented augmentation strategies. By evaluating multiple architectures in a unified framework, this study aims to identify the most effective and practical model for detecting modern deepfake images [5, 7–10].

The main objective of this paper is to design, develop, and evaluate a deep learning-based deepfake image detection system that can accurately distinguish manipulated images from authentic ones. In addition to comparing the performance of multiple models, the study also aims to select the most suitable model for deployment and integrate it into a user-friendly software system for practical use. The developed system allows users to upload an image and receive a prediction indicating whether the image is real or fake. This practical integration increases the real-world value of the research by linking model performance with usability and deployment potential [5, 6].

The main contributions of this study are as follows:

- A comparative evaluation of eight deep learning models for deepfake image detection under a consistent experimental setting.
- The use of a recent and diverse dataset to improve the relevance of the evaluation to modern deepfake generation techniques.
- The identification and optimisation of the most suitable model for deployment based on standard evaluation metrics such as accuracy, precision, recall, F1-score, and deployment considerations.
- The integration of the selected detection model into a practical web-based system for real-time image verification.
- A comparative-deployment evaluation framework that bridges offline model benchmarking with real-world software validation.

The remainder of this paper is organised as follows. Section 2 reviews existing studies on deepfake image detection. Section 3 describes the dataset, preprocessing steps, model architectures, and evaluation procedure. Section 4 presents the proposed detection system. Section 5 discusses the experimental results and comparative findings. Finally, Section 6 concludes the paper and outlines possible directions for future work.

2 Related Work

Deepfake image detection has become an important research area due to the rapid improvement of generative artificial intelligence techniques. As synthetic image generation models become more realistic, detection systems must also become more advanced to identify manipulated content accurately. Existing studies have proposed a wide range of deep learning approaches to detect deepfake images by analysing spatial artefacts, texture inconsistencies, frequency patterns, reconstruction errors, and multimodal relationships. This section reviews the major categories of deepfake image detection methods that are relevant to this study [2, 4].

2.1 Dense Convolutional Architectures

Dense convolutional neural networks are among the most widely used approaches for deepfake image detection. These models learn hierarchical spatial features from images and are effective in identifying subtle manipulation artefacts introduced during synthetic image generation. Patel et al. proposed a Deep Convolutional Neural Network (D-CNN) that uses multiple convolutional layers, batch normalization, and dropout regularization to classify images as real or fake. Their model showed strong performance on benchmark datasets by learning discriminative spatial features from manipulated facial images. Similarly, other dense convolutional models have used contrastive loss and dense connectivity to improve feature separation between authentic and manipulated content. These methods are effective in learning fine-grained visual inconsistencies, but they often show limited generalisation when evaluated on unseen datasets or on images generated using newer techniques. In practice, this means that strong benchmark accuracy does not always translate into reliable performance against newly emerging generators or compressed online images [7, 11].

2.2 Attention-Based Models

Attention-based methods have been developed to improve the ability of deepfake detectors to focus on important manipulated regions within an image. Unlike standard convolutional networks, attention mechanisms can assign greater importance to suspicious spatial regions or informative feature channels. Yang et al. introduced the Channel-Spatial-Triplet Attention Network (CSTAN), which combines dynamic convolution with channel and spatial attention to improve cross-dataset generalisation. This model uses attention modules to enhance the representation of subtle forgery traces and local artefacts that may be missed by ordinary convolution layers. Attention-based approaches have demonstrated better localisation of manipulated areas and improved robustness compared to conventional CNNs. However, these models are usually more complex and computationally expensive, which may limit their use in real-time applications. Their main strength is targeted feature extraction, but this advantage can diminish when deployment requires low latency or limited hardware resources [8].

2.3 Diffusion-Based Detection

Diffusion-based methods represent a more recent direction in deepfake detection research. Instead of relying only on spatial artefacts, these methods analyse the reconstruction behaviour of images using diffusion models. Ding et al. proposed the Deepfake Detection Enhanced Model (DDEM), which combines RGB and frequency-domain features with a diffusion reconstruction mechanism. Their method uses a denoising diffusion implicit model to reconstruct images and then measures the reconstruction error to identify possible manipulated regions. This approach allows both detection and localisation of deepfake content. Diffusion-based models are particularly useful for detecting advanced synthetic images because they capture statistical inconsistencies that are not always visible in the spatial domain. However, the reconstruction process requires high computational resources, making these methods less suitable for lightweight or real-time deployment [9].

2.4 Vision–Language Models

Vision–language models have recently been explored to improve the generalisation ability of deepfake detection systems. These methods combine visual and textual representations to learn stronger semantic differences between real and fake images. Tao et al. proposed the Language-Enhanced Deepfake Detection Network (LEDNet), which uses the CLIP model as a backbone for image and text encoding. Their framework integrates visual features with language-guided knowledge to strengthen the distinction between authentic and manipulated images. By aligning image representations with textual prompts such as “real photo” and “fake photo,” the model improves cross-dataset robustness and reduces overfitting to specific visual artefacts. Although vision–language methods show promising performance, they depend heavily on large pretrained models and require substantial memory and computational resources [5].

2.5 Ensemble Learning Approaches

Ensemble learning has been used in deepfake detection to improve classification reliability by combining the predictions of multiple models. Ben Jabra, Cheikhrouhou and BenAmor proposed an ensemble framework that integrates InceptionV3, VGG16, and Xception to classify images as real or fake. Their method aggregates model predictions using majority voting or weighted averaging, allowing the system to benefit from the strengths of different architectures. Ensemble models often produce better detection performance and greater robustness than single-model approaches. Some ensemble systems also incorporate interpretability techniques such as Grad-CAM to visualise the facial regions influencing the final prediction. Despite these advantages, ensemble methods increase computational cost and memory usage because multiple deep models must be executed together. Therefore, ensemble learning can improve robustness, but it is usually less suitable when inference speed is a critical requirement [6].

2.6 RAW-Domain Detection

Traditional deepfake detection methods usually operate on processed RGB images. However, image signal processing operations such as compression and colour correction can remove low-level forensic traces. To overcome this problem, some researchers have explored deepfake detection in the RAW sensor domain. Husseini and Dugelay proposed a RAW-domain detection framework that reconstructs RAW sensor information from RGB images using inverse image signal processing. Their framework also generates RAW self-blended images to simulate manipulation artefacts at the sensor level and trains an EfficientNet-B4 classifier for detection. RAW-domain methods preserve intrinsic camera characteristics and provide deeper forensic information than RGB-based approaches. Nevertheless, these models are limited by the lack of large-scale RAW datasets and the complexity of preprocessing. Their main value is forensic richness, but their practical adoption remains constrained by data availability and processing overhead [10].

Table 1: Comparative Summary of Deepfake Detection Approaches

Approach	Model	Technique	Contribution	Limitation
Dense CNN	D-CNN	CNN, BatchNorm, Dropout	Learns spatial features for deepfake detection	Poor generalization
Dense CNN	Dense-CNN	Contrastive Loss	Improves feature separation between real and fake images	Complex training
Attention	CSTAN	ODConv, Attention	Improves localization of manipulated regions	High computation
Diffusion	DDEM	DDIM, DIRE	Detects and localizes fake regions	Very expensive
Vision–Language	LEDNet	CLIP, LKA, AVM	Strong generalization across datasets	High memory usage
Ensemble	Ensemble	Voting, Averaging	Improves robustness and prediction stability	Slow and heavy
RAW Domain	RAW-SBI	Inverse-ISP	Enables sensor-level detection	Data limitation

2.7 Feature-Based and Image Quality Approaches

In addition to deep learning classifiers, some studies have explored feature-based methods for detecting manipulated facial images. These methods extract quality-related features from both frequency and spatial domains and use classical machine learning models for classification. Existing findings indicate that image quality measures can provide strong discrimination between real and fake faces and can achieve high performance on standard datasets. These approaches are attractive because they offer interpretability and lower complexity compared to large deep learning models. However, hand-crafted feature methods may be less adaptable to rapidly changing deepfake generation techniques when compared to modern data-driven deep learning systems.

2.8 Research Gap

Although many studies have reported strong performance for deepfake image detection, important challenges remain. Dense convolutional networks often perform well on familiar datasets but struggle to generalise to unseen manipulations. Attention-based and diffusion-based models improve robustness and localisation, but they introduce higher computational cost. Vision–language models offer improved semantic understanding, but they rely on large pretrained architectures. Ensemble systems improve reliability at the cost of efficiency, while RAW-domain approaches provide richer forensic information but face data availability and preprocessing limitations. These limitations show that no single method fully solves the deepfake detection problem in a way that is simultaneously accurate, efficient, and deployable.

More specifically, the literature still lacks two elements that are essential for practical progress. First, there is no sufficiently unified comparison of multiple major deepfake detection paradigms under the same dataset, preprocessing pipeline, training procedure, and evaluation criteria. To the best of this study's knowledge, existing work does not provide a single framework-level comparison of eight architecturally distinct deepfake detectors under one common protocol. Second, many studies stop at offline benchmark reporting and do not test how models behave inside a practical software system where latency, throughput, and user-facing errors matter. In addition, existing studies rarely explain how models that appear weaker offline may become more useful during deployment, or why models with high recall, such as RAW-domain approaches, may still be valuable despite lower overall accuracy. Therefore, the research gap is not only methodological but also operational: the field lacks a consistent benchmark that connects laboratory evaluation with real-world deployment behaviour. This study addresses that gap by implementing eight diverse detectors within a single framework and by validating them both in controlled experiments and in a usable software-based deepfake inspection system [2–4].

3 Methodology

This study adopts an experimental comparative methodology to evaluate the effectiveness of multiple deep learning approaches for deepfake image detection. The main purpose of the methodology is to identify a robust and practical model that can accurately distinguish manipulated images from authentic images generated in modern digital environments. To achieve this objective, the research process was organised into several stages, including dataset selection, data preprocessing, model implementation, training, evaluation, and system integration. A unified experimental setting was maintained across all models to ensure fair comparison of performance.

3.1 Research Design

The research follows an experimental comparative design in which different deep learning model architectures are trained and evaluated on the same dataset under controlled conditions. In this study, the independent variable is the model architecture used for deepfake detection, while the dependent variables are the performance outcomes measured using standard classification metrics. The controlled variables include the dataset, preprocessing pipeline, computational environment, and evaluation procedure. By keeping these factors consistent, the observed differences in performance can be attributed mainly to the capabilities of the selected architectures.

Eight deep learning models were implemented and compared in this research: D-CNN, CSTAN, OD-ResNet-34, DDEM, LEDNet, Ensemble Framework, RAW-SBI Pipeline, and SDAug. These models were selected because they represent different deepfake detection paradigms, including dense convolutional learning, attention-based architectures, diffusion-based detection, multimodal vision–language learning, ensemble classification, sensor-level forensic analysis, and augmentation-based robustness strategies. Evaluating these models within a single framework provides a broader understanding of their strengths and limitations for modern deepfake image detection.

3.2 Dataset

The dataset used in this study is the DeepDetect-2025 dataset, obtained from the Kaggle platform. This dataset was selected because it is recent, diverse, and specifically suitable for deepfake image detection research. The use of a modern dataset is important because older datasets may not represent the realism and diversity of current synthetic image generation techniques. The DeepDetect-2025 dataset contains both authentic and AI-generated fake images, allowing the models to learn meaningful differences between real and manipulated content.

The dataset is organised into training and testing subsets, and each subset is divided into two classes: real and fake. In total, the dataset contains 112,185 images. The training subset contains 48,815 real images and 41,594 fake images, while the testing subset contains 11,377 real images and 10,399 fake images. This relatively balanced class distribution reduces the risk of model bias toward one category and supports fair binary classification. The diversity of the dataset, including faces, objects, natural scenes, and urban environments, also improves the ability of the models to learn a wider range of visual manipulation artefacts.

3.3 Data Preprocessing

Before training, all images were preprocessed to ensure compatibility with the selected deep learning architectures and to improve training stability. Since the implemented models have different design requirements, the preprocessing pipeline was adapted according to each model. Input images were resized to match the expected resolution of each architecture, with image sizes ranging from 128×128

Table 2: DeepDetect-2025 Dataset Distribution

Subset	Real Images	Fake Images	Total Images
Training	48,815	41,594	90,409
Testing	11,377	10,399	21,776
Total	60,192	51,993	112,185

to 256×256 pixels. This resizing step ensured that the input data could be processed efficiently while preserving relevant visual information.

Image normalisation was applied to standardise pixel values and improve optimisation during training. Two normalisation strategies were used in this research. Some models applied standard normalisation using a mean value of 0.5, while others used ImageNet normalisation when transfer learning or pretrained backbones were involved. These choices were made according to the implementation requirements of the corresponding architectures.

Data augmentation was also used to improve generalisation and reduce overfitting. The augmentation strategy varied by model. Basic transformations such as horizontal flipping, rotation, and colour jittering were used in several models. Some architectures employed specialised preprocessing strategies. For example, the RAW-SBI Pipeline used self-blended image generation to simulate realistic manipulations, while SDAug introduced stochastic degradations such as JPEG compression, Gaussian blur, and noise to improve robustness under image distortions. In contrast, some models such as DDEM and OD-ResNet-34 were trained without augmentation.

Table 3: Preprocessing Configuration for Each Model

Model	Input Resolution	Normalization Strategy	Data Augmentation
DDEM	224×224	Standard (0.5)	None
OD-ResNet-34	160×160	ImageNet	None
CSTAN	128×128	Standard (0.5)	Horizontal Flip
D-CNN	160×160	Standard (0.5)	Horizontal Flip
LEDNet	256×256	Standard (0.5)	Horizontal Flip
Ensemble Framework	224×224	ImageNet	Horizontal Flip, Rotation, Color Jitter
RAW-SBI Pipeline	224×224	ImageNet	Horizontal Flip, Color Jitter, Self-Blended Images
SDAug	224×224	ImageNet	JPEG Compression, Gaussian Blur, Noise

3.4 Model Architectures

This study evaluates eight deep learning models representing different design philosophies for deepfake image detection. D-CNN is a custom dense convolutional neural network trained from scratch to capture spatial inconsistencies in manipulated images. CSTAN is implemented as a transfer learning based convolutional model that uses a pretrained ResNet backbone to extract hierarchical image features. OD-ResNet-34 extends residual learning with a stronger backbone design for image classification. DDEM is a diffusion-based detection framework that combines RGB and frequency-domain information with reconstruction error analysis. LEDNet is a vision–language based model that integrates semantic representations from visual and textual feature alignment. The Ensemble Framework combines multiple convolutional models to improve prediction reliability. RAW-SBI Pipeline operates at the reconstructed RAW image level to preserve low-level forensic traces, while SDAug focuses on robustness improvement through degradation-aware augmentation. Together, these models provide a representative sample of current deepfake detection strategies.

3.5 Training Procedure

All models were trained under a controlled computational setting to ensure fair comparison. The training process involved loading the preprocessed training images, applying the model-specific transformations, and optimising the model parameters using supervised learning. During training, each model was exposed to labelled real and fake images, allowing it to learn discriminative patterns that separate authentic images from manipulated ones. Mini-batch learning was used to improve memory efficiency and stabilise gradient updates. Optimisation was performed using common deep learning strategies such as the Adam optimiser and cross-entropy based objective functions, depending on the model implementation.

To reduce overfitting, regularisation strategies such as dropout, normalisation, augmentation, and transfer learning were employed where appropriate. Some models made use of pretrained weights to improve convergence and reduce training time, while others were trained from scratch to learn task-specific deepfake features directly from the dataset. The training process was repeated for all eight models, and their outputs were later compared using the same testing subset.

Table 4: Summary of Training Hyperparameters

Model	**Optimizer**	**Learning Rate**	**Batch Size**	**Input Size**
CSTAN	Adam	0.0001	32	128×128
D-CNN	Adam	0.001	32	160×160
DDEM	Adam	0.001	16	224×224
LEDNet	Adam	0.001	32	224×224
OD-ResNet-34	Adam	0.0001	32	160×160
RAW-SBI Pipeline	Adam	0.0002	16	224×224
SDAug	AdamW	0.0001	32	224×224
Ensemble Framework	AdamW	0.0001	16	224×224

3.6 Evaluation Metrics

Model performance was evaluated using standard binary classification metrics. Accuracy was used to measure the overall proportion of correctly classified images. Precision was used to determine the proportion of predicted fake images that were actually fake. Recall measured the ability of the model to identify fake images correctly, while the F1-score provided a balanced summary of precision and recall. These metrics are widely used in deepfake detection research because they offer a clear understanding of model reliability across both classes.

In addition to classification performance, confusion matrix analysis was used to examine the distribution of true positives, true negatives, false positives, and false negatives. This analysis helped identify how each model behaved under different prediction conditions. Computational performance was also considered in order to understand the practical efficiency of the architectures, particularly for real-world deployment scenarios.

3.7 System Integration

After the comparative evaluation, the most suitable model for deployment was selected and integrated into a practical software system. The purpose of this stage was to test the applicability of the selected model beyond experimental evaluation. The final system allows a user to upload an image and receive a prediction indicating whether the image is real or fake. This integration connects the research contribution to a usable application and demonstrates the practical value of the selected deepfake detection approach.

4 Proposed System

The proposed system in this study is a practical deepfake image detection platform developed to apply the most suitable model for deployment in a real-world usage scenario. After the comparative evaluation of multiple detection architectures, the selected model was integrated into a user-friendly software system. The purpose of this system is to provide a simple and accessible method for identifying whether an uploaded image is authentic or manipulated. By combining a trained deep learning model with a lightweight application interface, the proposed system extends the research contribution beyond experimental analysis and demonstrates its practical value for digital media verification.

4.1 System Overview

The proposed system follows a prediction pipeline that begins with image input and ends with classification output. A user uploads an image through the application interface, and the system processes the image using the same preprocessing configuration that was applied during model training. After preprocessing, the image is passed to the selected deep learning model, which extracts relevant features and performs binary classification. The final output is displayed to the user as a prediction label indicating whether the image is real or fake. This workflow ensures that the deployed system remains consistent with the experimental setup used during model development and evaluation.

4.2 System Architecture

The architecture of the proposed system consists of four main components: the user interface, the preprocessing module, the trained deep learning inference engine, and the output module. The user interface is responsible for receiving the uploaded image from the user. The preprocessing module then resizes and normalises the image according to the requirements of the selected model. The inference engine loads the trained model and applies it to the processed image to generate a classification result. Finally, the output module presents the prediction result to the user in a clear and understandable format. This modular design improves maintainability and allows the system to be extended or updated in future work.

4.3 Image Preprocessing in the Deployed System

To ensure compatibility between training and deployment, the uploaded image undergoes the same preprocessing operations used during the experimental stage. These operations include image resizing, pixel normalisation, and format conversion. The resizing step ensures that the image matches the required input resolution of the selected model. Normalisation standardises the image data and supports stable inference. In cases where the selected model relies on a transfer learning backbone or a special input format, the deployed preprocessing pipeline is aligned with that requirement. This consistency is important because mismatches between training-time and deployment-time preprocessing can reduce detection accuracy.

4.4 Deep Learning Inference Module

The core of the proposed system is the inference module, which contains the trained deep learning model selected from the comparative analysis. This module receives the preprocessed image and performs feature extraction and classification. During inference, the model analyses visual patterns, manipulation artefacts, texture inconsistencies, and other discriminative clues learned during training. The output is a probability-based prediction or a final binary decision indicating whether the image is authentic or manipulated. Since the selected model was chosen based on comparative performance and deployment suitability, this module represents the most appropriate detection strategy identified in the study.

4.5 Output and User Interaction

Once inference is complete, the system returns the result through the application interface. The user is shown a simple prediction such as *Real* or *Fake*. Depending on the implementation, the system may also provide a confidence score to indicate the certainty of the prediction. Presenting the result in a clear and direct way improves usability for non-technical users and supports real-world applicability. The interface is designed to minimise complexity, allowing users to interact with the system without requiring knowledge of deep learning or digital forensics.

4.6 Real-World Validation of the Software System

To assess the practical behaviour of the deployed Deepfake Inspector software, the integrated models were also evaluated in a real-world usage setting. The system supports batch validation by allowing the user to upload real-image and fake-image sets so that model behaviour can be checked directly through the application interface. This validation considered not only classification accuracy but also user-facing performance factors such as inference speed, average latency, and the distribution of false positive and false negative predictions. These measures are important because a deployable deepfake detection system must support both reliable decisions and responsive interaction.

The real-world validation results showed that model behaviour in the deployed software environment differed from the controlled experimental setting. The real-world validation was conducted on a

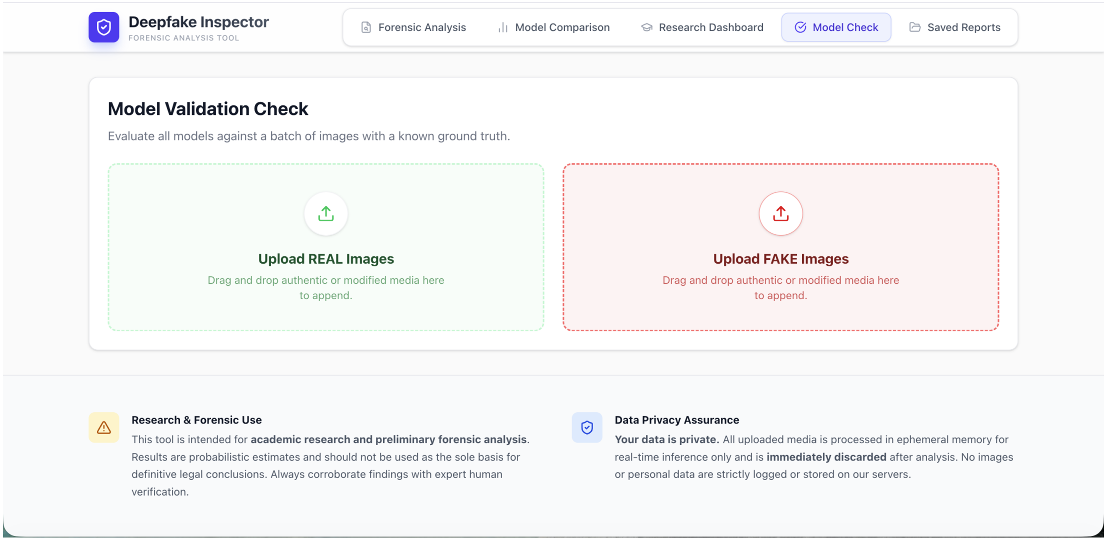

Fig. 1: Interface used for real-world validation of model predictions in the Deepfake Inspector software.

Table 5: Real-World Validation Results from the Deepfake Inspector Software System

Model	Accuracy	FPS	Avg Latency	Correct	False Positives	False Negatives
DDEM	75.0%	8.0	125 ms	33/44	9	2
D-CNN	61.4%	55.6	18 ms	27/44	4	13
SDAug	56.8%	8.1	124 ms	25/44	14	5
CSTAN	52.3%	71.4	14 ms	23/44	8	13
OD-ResNet-34	52.3%	19.6	51 ms	23/44	3	18
Ensemble	50.0%	2.6	381 ms	22/44	1	21
LEDNet	45.5%	142.9	7 ms	20/44	4	20
RAW-SBI Pipeline	43.2%	5.7	176 ms	19/44	3	22

limited batch of samples; therefore, the observed performance should be interpreted as preliminary deployment evidence rather than a fully generalised evaluation. DDEM achieved the highest software-level validation accuracy, while D-CNN and CSTAN offered faster interactive performance. Some models produced fewer false positives but missed more manipulated images, whereas others improved detection sensitivity at the cost of additional false alarms. These findings reinforce the importance of evaluating models not only through offline test metrics but also through practical software-based validation.

4.7 Practical Significance of the System

The development of the proposed system demonstrates that deepfake detection research can be translated into a usable application. While many research studies focus only on model performance, this work also considers deployment and usability. The system can support practical image verification in situations where trust in digital content is important, such as online communication, media sharing, and cybersecurity awareness. By integrating the most suitable model for deployment into a deployable system, the study contributes not only to comparative research but also to the practical adoption of deepfake image detection technology.

4.8 System Limitations

Although the proposed system provides a practical solution for image-based deepfake detection, it has several limitations. First, the system focuses only on still images and does not currently handle video deepfakes or multimodal manipulations involving both image and audio. Second, the model performance depends on the quality and diversity of the training dataset, which means that highly novel manipulation methods may still reduce detection reliability. Third, some advanced architectures may require considerable computational resources, which can affect real-time performance in limited

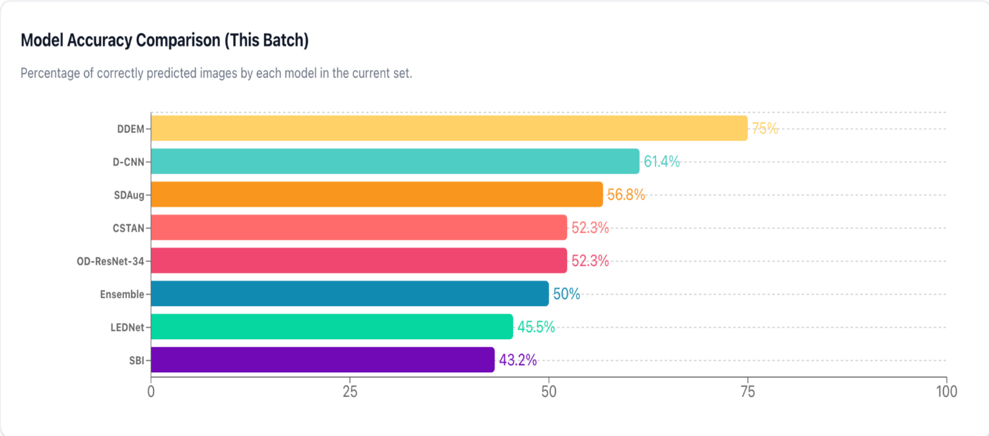

Fig. 2: Model accuracy comparison for the latest real-world validation batch executed through the Deepfake Inspector software system.

hardware environments. These limitations highlight the need for continued improvement in future research and deployment.

5 Results and Discussion

This section presents the experimental findings obtained from the comparative evaluation of the eight deep learning models implemented in this study. The purpose of this analysis is to determine how effectively each model can distinguish real images from deepfake images under the same dataset and evaluation setting. The discussion focuses on classification performance, comparative behaviour, and practical observations related to model selection for deployment.

5.1 Overall Model Performance

The eight implemented models, namely D-CNN, CSTAN, OD-ResNet-34, DDEM, LEDNet, Ensemble Framework, RAW-SBI Pipeline, and SDAug, were evaluated using standard binary classification metrics. These metrics included accuracy, precision, recall, and F1-score. The results showed that performance varied across the different architectural approaches, indicating that deepfake image detection effectiveness depends strongly on the design of the model and the representation strategy it uses. Models that combined strong feature extraction with improved robustness generally achieved better performance than simpler baselines.

In general, architectures designed to capture richer manipulation traces or improve generalisation performed better than models that relied mainly on standard spatial feature learning. Attention-based, augmentation-enhanced, and ensemble-driven methods demonstrated stronger ability to recognise subtle fake-image patterns. At the same time, some computationally advanced methods showed practical limitations despite their strong detection capability. These findings confirm that high classification accuracy alone is not sufficient when selecting a model for real-world use; computational efficiency and generalisation are also important.

5.2 Classification Metrics

Accuracy was used to measure the overall proportion of correctly classified images. Precision was used to evaluate the reliability of fake-image predictions, while recall measured the ability of the system to correctly identify manipulated images. The F1-score provided a balanced assessment of both precision and recall. These metrics were selected because deepfake detection requires not only correct overall classification but also strong sensitivity to manipulated content, especially in security-related applications.

The comparative results showed that some models achieved high accuracy but lower recall, indicating that they were conservative in identifying fake images. Other models achieved stronger recall but at the cost of lower precision, meaning that they produced more false alarms. Therefore, the F1-score was especially useful for identifying models with balanced performance. The most suitable model for deployment was selected by considering the combination of these metrics rather than depending on a single score.

Table 6: Classification Performance of the Evaluated Models

Model	Accuracy	Precision	Recall	F1-score	AUC
CSTAN	82.11	85.87	82.11	81.78	93.48
SDAug	88.80	90.77	88.80	88.66	98.51
D-CNN	88.41	97.42	79.93	87.81	98.14
LEDNet	78.42	93.92	62.75	75.23	95.19
RAW-SBI Pipeline	58.76	53.74	97.89	69.39	78.26
DDEM	53.12	50.46	45.15	47.81	98.35
OD-ResNet-34	73.10	80.41	65.79	69.45	78.86
Ensemble	45.10	48.50	81.80	35.60	49.61

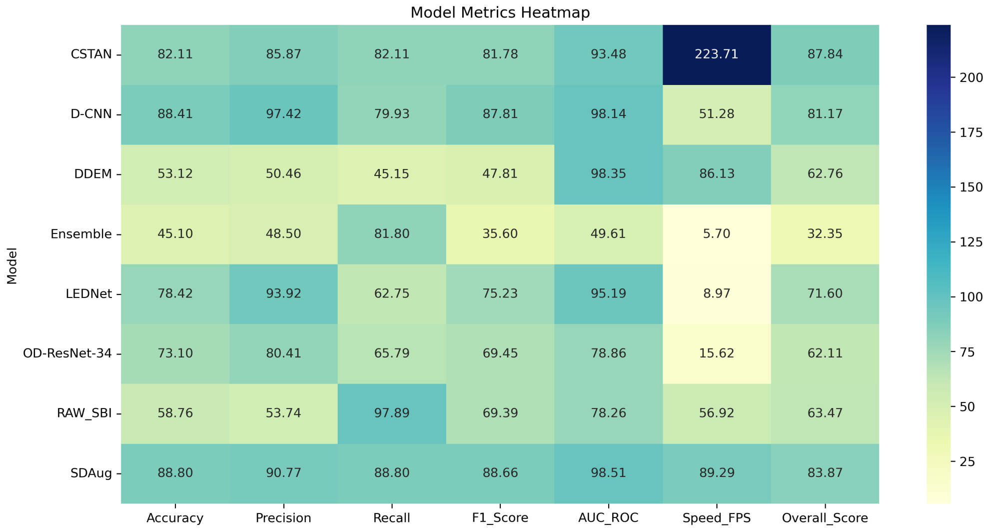

Fig. 3: Metrics heatmap summarising the comparative performance of the evaluated models across the key measures of accuracy, F1-score, AUC, FPS, and overall score.

Table 7: Computational Performance of the Evaluated Models

Model	Inf. Time (ms)	Model Size (MB)	FPS	Overall Score
CSTAN	4.47	42.64	223.71	87.84
SDAug	11.20	17.95	89.29	83.87
D-CNN	19.50	4.06	51.28	81.17
LEDNet	111.48	0.72	8.97	71.60
RAW-SBI Pipeline	17.57	31.19	56.92	63.47
DDEM	11.61	15.58	86.13	62.76
OD-ResNet-34	64.04	163.63	15.62	62.11
Ensemble	175.53	448.41	5.70	32.35

5.3 Confusion Matrix Analysis

Confusion matrix analysis was used to examine how each model performed across the two classes, real and fake. This analysis provided insight into the number of true positives, true negatives, false positives, and false negatives produced by each architecture. A strong deepfake detector should not only classify authentic images correctly but should also minimise the number of manipulated images that are incorrectly predicted as real. False negatives are particularly important in this context because undetected fake images can create security and trust-related risks.

The confusion matrix results showed that certain models were more effective at identifying fake images, while others performed better at preserving correct classification of real images. This indicates a trade-off between sensitivity and specificity across model architectures. Models with better-balanced confusion matrices were considered more practical because they reduced the risk of both missed detections and unnecessary false warnings.

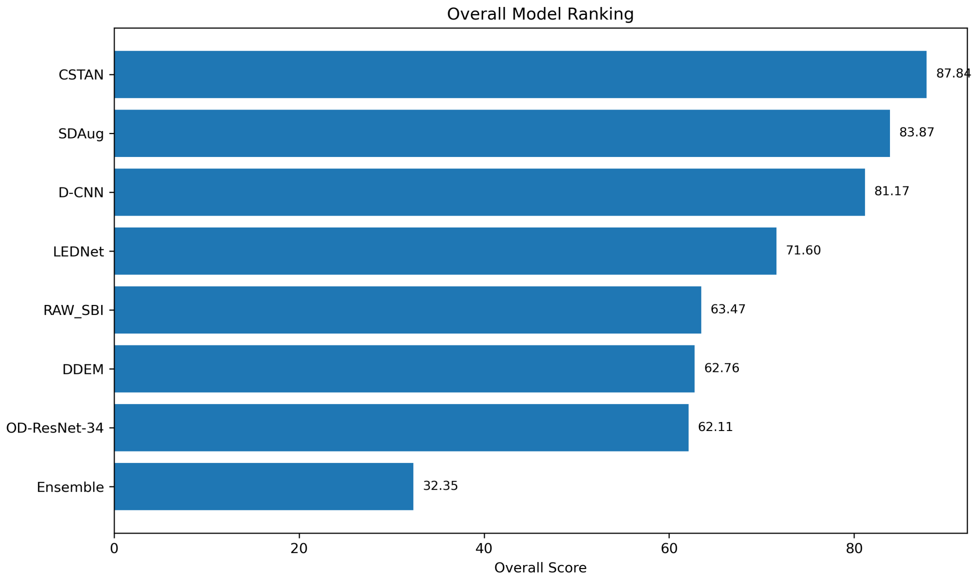

Fig. 4: Overall ranking of the evaluated deepfake detection models based on their combined comparative performance.

5.4 Comparative Discussion of Model Behaviour

The D-CNN model provided a strong baseline by learning spatial manipulation artefacts directly from the dataset, but its performance was limited by reduced generalisation compared to more advanced approaches. CSTAN and OD-ResNet-34 improved representation learning through stronger backbone structures, making them more effective at capturing discriminative image patterns. DDEM introduced reconstruction-based detection, which improved its ability to identify subtle forgery traces, although this came with higher computational complexity. LEDNet demonstrated the value of multimodal semantic guidance, which helped improve robustness across image types. The Ensemble Framework benefited from combining multiple convolutional architectures, resulting in improved reliability, while the RAW-SBI Pipeline preserved low-level forensic cues that are often lost in standard RGB processing. SDAug improved robustness by training under degraded image conditions, making it more resistant to distortions such as compression and blur.

These observations suggest that no single architectural strategy is universally superior in all aspects. Instead, each model offers a different balance between accuracy, robustness, generalisation, and efficiency. The model selected for system integration was the one that achieved the most suitable balance across these dimensions.

5.5 Computational Performance and Practical Considerations

In addition to classification quality, practical deployment requires attention to computational efficiency. Models such as ensemble frameworks, diffusion-based methods, and large multimodal systems can produce strong results, but they often require higher memory usage and longer inference time. This can make deployment difficult in environments with limited hardware resources or in applications that require near real-time prediction. In contrast, simpler or moderately complex architectures may offer slightly lower performance but better deployment feasibility.

For this reason, the selection of the final model was not based only on the highest detection metric. The practical goal of the research also required the chosen model to be suitable for integration into a user-facing system. Therefore, both predictive performance and operational efficiency were considered in identifying the final deployed model.

5.6 Discussion of Research Findings

The results of this study show that deepfake image detection benefits from comparative evaluation across multiple architectural families. Models that include attention mechanisms, advanced augmentation, multimodal guidance, or richer forensic representations generally offer stronger performance than standard CNN baselines. However, these improvements often introduce greater complexity. The

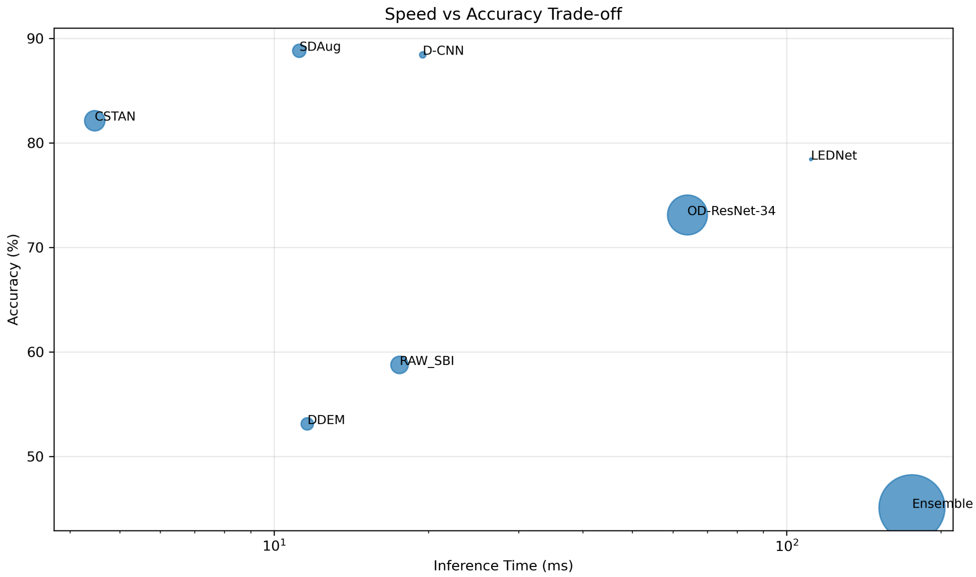

Fig. 5: Speed versus accuracy trade-off among the evaluated models, highlighting the balance between predictive quality and runtime efficiency.

findings also show that training and testing on a recent and diverse dataset is essential for evaluating the real usefulness of a model against modern deepfake generation techniques.

Several deeper observations emerge from the comparison. First, SDAug achieved the strongest overall balance because its stochastic degradation strategy exposed the model to compression, blur, and noise during training. This likely improved robustness under realistic image conditions rather than only improving benchmark accuracy. Its strong performance therefore appears to come from learning manipulation cues that remain stable under degraded acquisition and sharing conditions. Second, DDEM presented an important contrast between offline evaluation and software-level validation. Although it was not the top model in the comparative metrics table, it achieved the strongest real-world validation accuracy in the deployed system. This suggests that reconstruction-based detection may preserve useful forensic cues under practical operating conditions that are not fully captured by standard offline ranking. However, because the software validation batch was limited in size, this observation should be interpreted as preliminary deployment evidence rather than final generalisation proof. Third, D-CNN remained competitive because it combined relatively strong predictive performance with fast inference, showing that simpler architectures may still be preferable when deployment efficiency is critical. Fourth, RAW-SBI Pipeline produced the highest recall, which indicates strong sensitivity to manipulated samples. Although its overall accuracy was lower, its ability to detect a large proportion of fake images suggests that sensor-level forensic traces can be especially valuable in security-focused settings where missed detections are more costly than false alarms.

Compared with prior studies, the results obtained here are broadly consistent with the literature in showing that attention-enhanced and robustness-oriented detectors tend to outperform simpler baselines, while multimodal and reconstruction-based methods often offer different trade-offs between accuracy and deployment cost. For example, CSTAN, LEDNet, DDEM, and recent comparative assessments all report that robustness and cross-dataset behaviour improve when models capture richer forensic cues or semantic context, but these gains usually come with higher complexity or weaker runtime efficiency [3–5, 8, 9]. In that sense, the present results are less important as direct score matching against individual published experiments and more useful as a controlled comparison showing how the relative ranking of detector families changes when classification quality, computational efficiency, and software-level validation are considered together.

These findings indicate that deepfake detectors should not be judged only by a single aggregate score. Their practical value depends on how well they balance accuracy, robustness, inference speed, sensitivity to fake content, and behaviour under deployment conditions. In this sense, the main insight of this study is not only which model performs best, but also how the relative value of different detector families changes when evaluation moves from benchmark comparison to real software use. This comparative-deployment perspective constitutes the main practical novelty of the work, because it shows how model selection changes once offline metrics are interpreted together with software-level validation.

The results should also be interpreted with several experimental limitations in mind. The dataset, although recent and diverse, may still contain source-specific biases that favour some detector families over others. In addition, the software validation stage used a limited batch of samples, which restricts the strength of conclusions about deployment-time generalisation. Finally, because all models were evaluated within one unified implementation setting, the findings provide a fair internal comparison, but they do not replace large-scale cross-dataset benchmarking against every condition reported in the wider literature.

Overall, the experimental findings support the view that effective deepfake detection requires a balance between feature learning strength, robustness to unseen manipulations, and deployment practicality. This study contributes to the field by comparing several major detection paradigms under one unified framework and by using the findings to guide the development of a practical deepfake image detection system.

6 Conclusion

This study presented a comparative deep learning approach for deepfake image detection using a unified experimental framework. The research focused on evaluating multiple modern detection architectures in order to identify an effective and practical solution for distinguishing manipulated images from authentic images. To achieve this goal, eight deep learning models representing different detection paradigms were implemented and compared using a recent and diverse dataset. These models included dense convolutional architectures, attention-based methods, diffusion-based detection, vision–language learning, ensemble classification, RAW-domain analysis, and augmentation-based robustness strategies.

The experimental results showed that deepfake detection performance depends strongly on model design, feature representation strategy, and robustness to modern manipulation techniques. While some models demonstrated strong accuracy, others provided better balance across precision, recall, and F1-score. The findings confirmed that more advanced approaches generally outperform simple baseline models in detecting subtle manipulation artefacts. However, the study also showed that high predictive performance must be considered together with computational efficiency and deployment practicality, especially when the final goal is to integrate the selected model into a usable software system.

Based on the comparative evaluation, the most suitable model for deployment was selected and integrated into a practical image verification system. This integration demonstrated that deepfake detection research can move beyond theoretical analysis and be applied in a real-world setting where users can upload images and receive detection results. In this way, the study contributes not only to comparative research on deepfake detection architectures but also to the development of a usable detection application that supports digital trust and cybersecurity.

Overall, this research highlights the importance of evaluating multiple deepfake detection approaches under consistent conditions. This work provides a unified benchmark and practical deployment framework that connects experimental evaluation with real-world deepfake detection performance. It also shows that recent datasets, robust preprocessing, balanced evaluation metrics, and practical deployment considerations are all important for building reliable detection systems. The study provides a useful foundation for future work aimed at improving the generalisation, efficiency, and real-world applicability of deepfake image detection technologies. Its principal contribution is not a new detection architecture, but a clearer evidence-based understanding of how different detector families behave when the evaluation moves from controlled experiments to real deployment conditions.

7 Future Work

Future research can extend this study in several directions. One important improvement is to expand the system from image-based deepfake detection to video-based detection, where temporal inconsistencies can also be analysed. Another direction is to evaluate the selected models on larger and more diverse datasets that include newly emerging generative techniques. Future work may also focus on reducing computational complexity so that strong detection models can be deployed on resource-constrained devices. In addition, multimodal detection approaches that combine image, video, and audio analysis may provide stronger protection against more advanced deepfake attacks.

These improvements would help develop more generalisable, efficient, and practical deepfake detection systems for real-world applications.

Data Availability

The dataset used in this study is available from the corresponding author upon reasonable request, subject to dataset licensing terms and institutional requirements.

Declarations

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Conflict of Interest: The authors declare that they have no conflict of interest.

Author Contributions: V. Lambodaran led the implementation, experiments, system development, and manuscript drafting. K. Thiruthanigesan supervised the study, contributed to the research design, and reviewed the manuscript.

Ethics Approval: This study did not involve human participants, animal subjects, or clinical data.

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